Paper 67 — Beyond the Edge of Stability via Two-step Gradient Updates

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The paper in one minute

Question: Why can GD with a step size beyond the usual stability threshold remain bounded and settle into oscillatory training?
Model/Method: Analyze the *two-step* GD map, higher derivatives around minima, quartic toy losses, teacher–student networks and matrix factorization.
Finding: Under explicit local derivative conditions, the two-step map can have stable fixed points representing a period-two GD orbit; selected models converge to these cycles.
Why we care: Convergence of *two-step behavior* need not mean that individual iterates settle at a stationary point.

1. What changes beyond the edge of stability?

A standard quadratic analysis says the minimizer becomes unstable when $\eta f''(x^*) > 2$. But a nonlinear loss can flatten on either side of its minimizer. Then GD can bounce across it, with the curvature on successive sides compensating. A trajectory may approach a *two-cycle*, moving $a \to b \to a$, rather than a single parameter value.
Write one step as $F_\eta(x) = x - \eta f'(x)$; two steps give $F_\eta^2(x) = F_\eta(F_\eta(x))$. If $F_\eta^2(a) = a$ but $F_\eta(a) = b \neq a$, the original GD iteration oscillates forever between a and b. That is not the same as divergence or convergence to the minimizer.

2. Worked example: a quartic with a period-two opportunity

Let

$$f(x) = \tfrac{1}{4}(x^2 - 1)^2, \quad f'(x) = x(x^2 - 1), \quad f''(x) = 3x^2 - 1.$$

At the minimum $x^* = 1$, $f''(1) = 2$, so the quadratic edge is $\eta = 2/2 = 1$. Choose $\eta = 1.05$ and $x_0 = 0.8$. Then

$$x_1 = 0.8 - 1.05\,[0.8(0.64 - 1)] = 1.1024.$$

The next step is $x_2 = 1.1024 - 1.05[1.1024(1.1024^2 - 1)] \approx 0.8532$. The iterates cross the minimizer, but the unequal curvatures on the two sides need not make the cycle explode.
Important: Two displayed iterates illustrate the nonlinear bounce; they do *not* by themselves prove a stable orbit. Theorem 2 supplies a rigorous period-two convergence result for this quartic family in a stated step-size range.

3. Why study the second iterate?

For a general one-dimensional smooth loss with a local minimizer $\bar{x}$, Theorem 1 identifies the higher-order condition

$$f^{(3)}(\bar{x}) \neq 0, \qquad 3[f^{(3)}(\bar{x})]^2 - f''(\bar{x}) f^{(4)}(\bar{x}) > 0.$$

Under its local hypotheses and a suitable step slightly *above* $2/f''(\bar{x})$, two-step fixed points can exist. The derivative conditions measure how the curvature *changes* as the iterate crosses the minimum; constant-curvature quadratics have no such stabilizing mechanism.

Check the worked quartic

For $f(x) = \frac{1}{4}(x^2-1)^2$, at $x = 1$ we have $f'' = 2$, $f^{(3)} = 6$, $f^{(4)} = 6$, so $3(6)^2 - 2(6) = 96 > 0$. The quartic therefore has the right nonlinear geometry for the condition. Theorem 2 is the paper's stronger convergence statement for this model.

4. Results to distinguish

Theorem 1: existence of fixed points of the two-step map under derivative conditions in 1-D.
Theorem 2: for a special quartic $\frac{1}{4}(x^2-\mu)^2$, $\eta = K/\mu$ with $1 < K < \sqrt{4.5} - 1 \approx 1.121$, and specified initialization, GD converges to a period-two orbit (apart from exceptional starts).
Theorem 3 / Proposition 3: a related bilinear teacher–student loss, $\frac{1}{2}(xy-\mu)^2$, can reduce toward a balanced situation and display such a cycle under additional assumptions.
Further evidence: Results in one-neuron models and matrix factorization show oscillatory behavior in more dimensions, but not a universal theorem for deep neural networks.

5. Stability of the two-step orbit

A useful calculation is

$$(F_\eta^2)'(a) = F_\eta'(b) F_\eta'(a) = \big[1 - \eta f''(b)\big]\big[1 - \eta f''(a)\big], \qquad b = F_\eta(a).$$

The pair (a, b) can be locally attractive when the absolute value of this *product* is below one, even if one step expands. That is why nonlinear curvature on both sides of a minimizer matters. For a pure quadratic, $f''(a) = f''(b) = \kappa$ and the product becomes $(1-\eta\kappa)^2 > 1$ beyond EoS; there is no compensation.
The results in Sections 5–6 apply this idea to specific neural and matrix-factorization settings; their hypotheses are not identical to the one-dimensional quartic theorem.

5. Assumptions and limitations

Assumptions: Deterministic fixed-step GD, smooth local derivatives, specific higher-order curvature signs, special initialization and learning-rate intervals for the global quartic theorem; population-loss assumptions in the teacher–student examples.
Limitations: A stable two-cycle is not convergence of parameters to a minimizer. Proofs in low-dimensional examples do not imply that arbitrary networks exhibit stable oscillation above EoS; higher-dimensional observations are more exploratory. Some results are existence statements rather than attraction-basin characterizations.

6. Connections to the survey

Li–Wang–Li (2022) analyzes how sharpness moves around EoS; this paper asks what *orbit* GD approaches beyond it. **Kalra et al. (2025)** connects increased learning rates to period doubling and chaos in a simplified UV model. **Galli (2026)** studies adaptive nonmonotone steps but proves a different first-order guarantee.

7. Quick reference

Concept	Meaning / equation
One-step map	$F_\eta(x) = x - \eta f\prime(x)$
Two-step fixed point	$F_\eta^2(a) = a$, but $F_\eta(a) \neq a$
Local condition	$3(f^{(3)})^2 - f\prime\prime f^{(4)} > 0$
Quartic condition	$1 < K < \sqrt{4.5} - 1$
Original to read	Sections 4–6; Theorems 1–3

8. Papers Master note

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Proves existence of stable period-two GD behavior beyond classical quadratic EoS in selected smooth low-dimensional models, using two-step maps and higher derivatives. Do not equate two-cycle convergence with parameter convergence or universal deep-network stability.

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